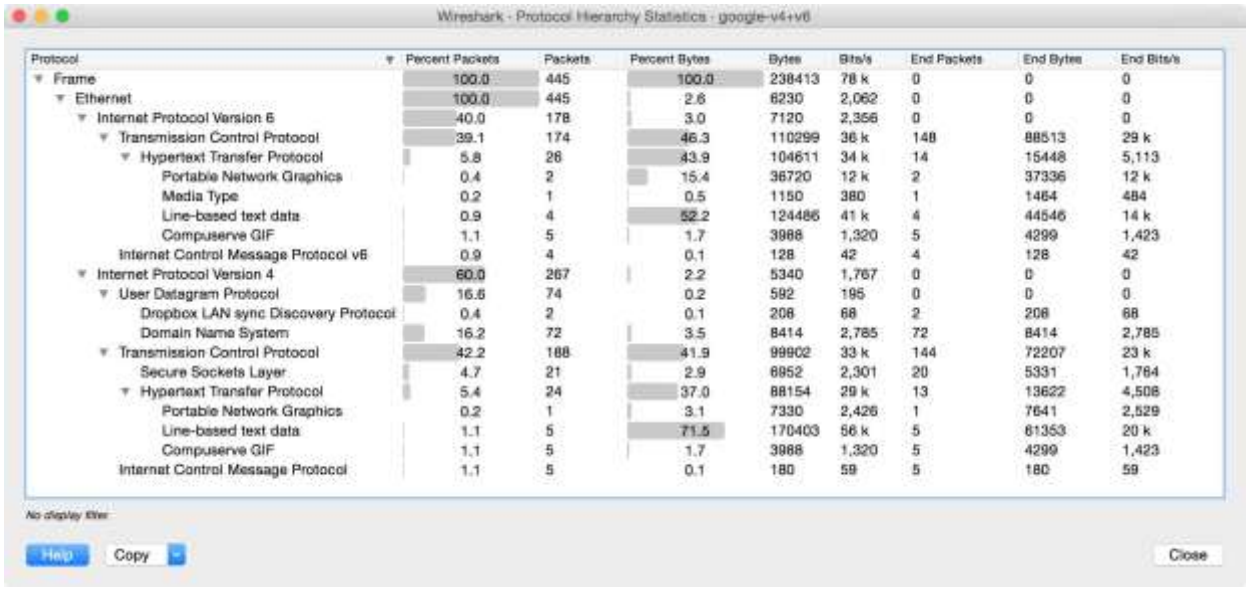
Software and Hardware Tools

Important: tasks 1 to 4 must be done on your own laptop and network, so I cannot capture the results for you. Follow the steps, paste your own screenshots in the boxes and fill in the blanks. The 'sample' images only show the format; their numbers and names are made up.

Task 1: Wireshark Capture (2 minutes)

1. Install Wireshark from wireshark.org (and Npcap if asked) and open it.
2. Double-click your active network (Wi-Fi) and capture starts. Browse Instagram or YouTube for 2 minutes.
3. Click the red Stop button, then File > Save As and save as a .pcap file (for example instagram_capture.pcap).
4. Open Statistics > Protocol Hierarchy and note the 3 protocols with the highest % of packets.

Your screenshot (paste your Protocol Hierarchy screenshot here):



Item	Your answer
Saved file name	_____.pcap
Top protocol 1	
Top protocol 2	
Top protocol 3	

Sample (example format only):

Wireshark (SAMPLE - example format only)

```
Protocol Hierarchy (top protocols by packets)

Protocol      % Packets   Packets
TCP              62.0%      3100
  TLS            48.5%      2425
UDP              33.0%      1650
  QUIC           28.0%      1400
  DNS             4.2%       210
ARP               1.1%        55

Saved as: instagram_capture.pcap
```

Note: on Instagram or YouTube you will usually see TCP, TLS (encrypted web traffic), UDP/QUIC and DNS near the top, but your own capture decides your answer. A capture can contain private data, so do not share the .pcap file publicly.

Task 2: ping Spotify and Zomato

Commands to run:

```
ping www.spotify.com
ping www.zomato.com
```

Your screenshot (paste both ping results here):



Website	Minimum (ms)	Average (ms)	Maximum (ms)
www.spotify.com			
www.zomato.com			

The faster site (lower average time): _____

Sample (example format only):

Command Prompt (SAMPLE - example format only)

```
C:\> ping www.spotify.com
Reply from 198.51.100.77: bytes=32 time=31ms TTL=57
Reply from 198.51.100.77: bytes=32 time=30ms TTL=57
Reply from 198.51.100.77: bytes=32 time=32ms TTL=57
Reply from 198.51.100.77: bytes=32 time=31ms TTL=57
    Minimum = 30ms, Maximum = 32ms, Average = 31ms

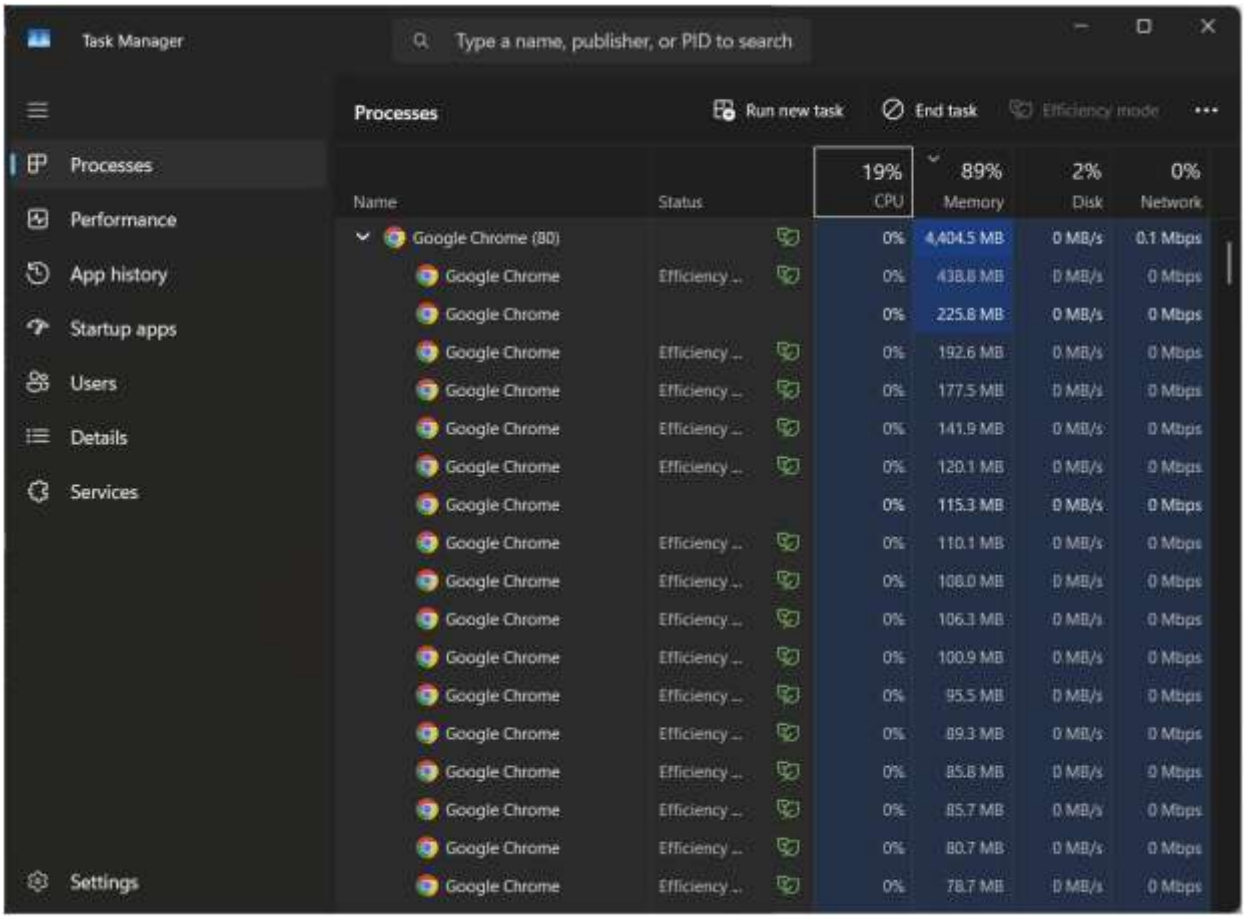
C:\> ping www.zomato.com
Reply from 203.0.113.50: bytes=32 time=18ms TTL=58
Reply from 203.0.113.50: bytes=32 time=19ms TTL=58
Reply from 203.0.113.50: bytes=32 time=17ms TTL=58
Reply from 203.0.113.50: bytes=32 time=18ms TTL=58
    Minimum = 17ms, Maximum = 19ms, Average = 18ms
```

Note: if a site shows 'Request timed out', it may simply block ping. In that case run the test again or use tracert/traceroute, and mention it in your answer.

Task 3: Running Processes in Task Manager / Activity Monitor

1. Windows: press Ctrl + Shift + Esc, open the Details tab (or Processes tab) and sort by Name. Mac: open Activity Monitor and use the Memory tab.
2. Find all processes for Chrome (or your browser) or WhatsApp Desktop and note the name and the memory used.

Your screenshot (paste your Task Manager screenshot here):



Sample (example format only):

Task Manager > Details (SAMPLE - example format only)

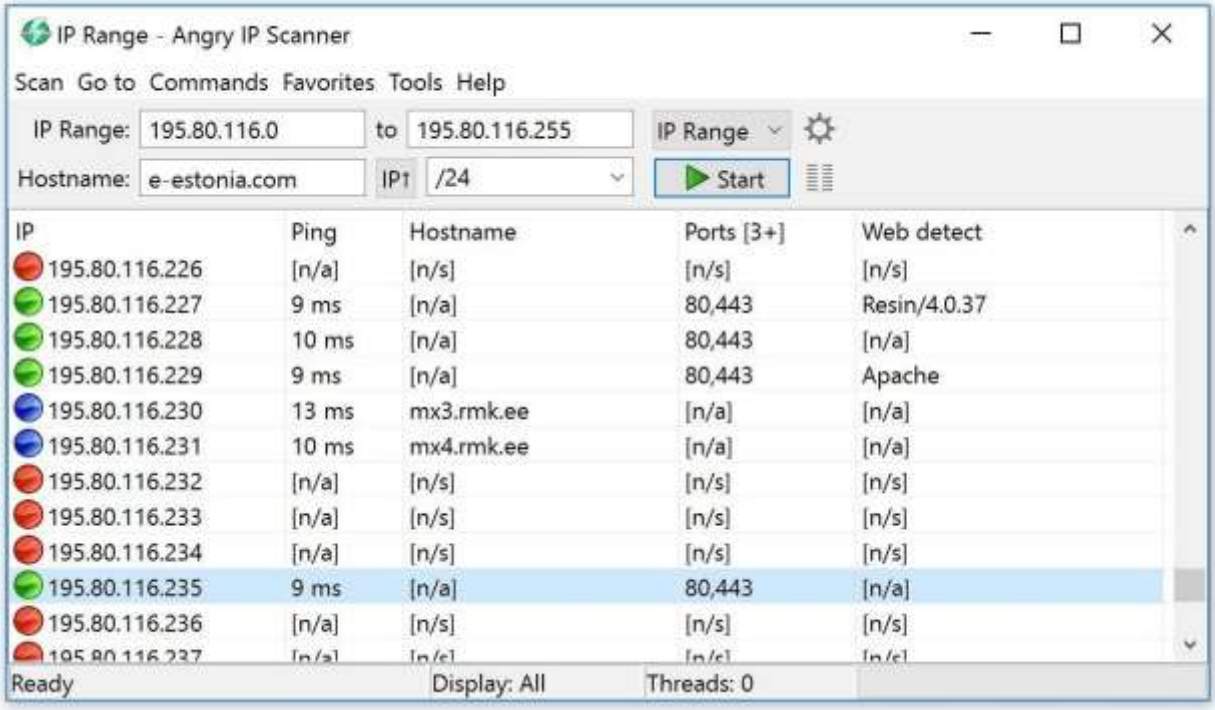
```
Process name              Memory
chrome.exe                 185 MB
chrome.exe (renderer)      240 MB
chrome.exe (GPU process)    95 MB
chrome.exe (utility)        40 MB
WhatsApp.exe               150 MB
```

Note: Chrome shows many processes because each tab, extension and the GPU run separately for stability and security.

Task 4: Angry IP Scanner (your own network only)

1. Download Angry IP Scanner from angryip.org and install or run it.
2. Check that the IP range is your own network (for example 192.168.1.1 to 192.168.1.254), then click Start. Scan only your home or college network.
3. When the scan finishes, note at least 2 devices with their IP and hostname (device name).

Your screenshot (paste your scan result here):



Sample (example format only):

Angry IP Scanner result (SAMPLE - example format only)

```
IP              Hostname / Device name
192.168.1.1     router.home
192.168.1.8     LAPTOP-STUDENT
192.168.1.12    Galaxy-Phone
192.168.1.15    Smart-TV
```

Task 5: Three Hardware Tools for Network Troubleshooting

Note: this list was written by Claude (an AI assistant), not ChatGPT or Bing AI. If your assignment requires one of those, ask it the same question and compare the answers.

Hardware tool	When you would use it	Example
Cable tester	When a computer shows 'network cable unplugged' or has a slow or broken wired connection, to check that every wire inside the LAN cable is connected correctly.	A lab PC will not connect to the switch; the tester shows one wire is broken in the patch cable.
Crimping tool (with RJ45 connectors)	When you must make a new LAN cable or replace a damaged connector on a cable end.	A cable end is cracked, so you cut it and crimp a fresh RJ45 connector.
Multimeter	When you suspect a power problem, to measure voltage or check continuity of a cable or power adapter.	A router will not switch on; the multimeter shows its adapter gives no output voltage.